Multiply-Upstream, Semi-Lagrangian Advective Schemes: Analysis and Application to a Multi-Level Primitive Equation Model

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ABSTRACT

The stability properties of some simple semi-Lagrangian advective schemes, based on a multiply-upstream interpolation, are examined. In these schemes, the interpolation points are chosen to surround the departure points of the fluid particles at the beginning of a time step. It is shown that the schemes, though explicit, are unconditionally stable for a constant wind field.

Application of the schemes to a multi-level split explicit model shows that they enable full advantage to be taken of the splitting method by allowing a long time step for advection. It is shown that they can thus lead to a considerable saving of computer time compared to Eulerian schemes, while giving comparable accuracy.

1. Introduction

A Lagrangian approach to solving the equations of fluid motion involves following a fixed set of particles throughout the period of integration. Such an approach was first used in meteorology by Fjørtoft (1952) when he proposed a graphical method of solving the barotropic vorticity equation using a single time step of 24 h. However, Welander (1955) showed that in general a set of fluid particles which are initially regularly distributed soon become greatly deformed and are thus rendered unsuitable for numerical integration. To avoid this difficulty, while still concentrating on fluid particles, Wiin-Nielsen (1959) introduced a semi-Lagrangian (also referred to as quasi-Lagrangian) approach, whereby a set of particles which arrive at a regular set of grid points are traced backwards over a single time interval to their departure points. The values of the dynamical quantities at the departure points are obtained by interpolation from neighboring grid points where their values are known. The semi-Lagrangian differs from the Lagrangian approach because in the former the set of particles in question changes at each time step.

The semi-Lagrangian approach has been used in numerical prediction models by a number of authors (Krishnamurti, 1962, 1969; Økland, 1962; Sawyer, 1963; Leith, 1965; Mahrer and Pielke, 1978). Analyses of the properties of semi-Lagrangian advective schemes have also been given by Crowley (1968), Mathur (1970), Purnell (1976) and Long and Pepper (1981).

Only recently, however, with the work of Robert (1981), has it been clearly demonstrated that the use of the semi-Lagrangian approach may offer significant advantages over the purely Eulerian approach for numerical weather prediction. Robert has shown that it is possible to perform stable integrations of a divergent barotropic model with normal spatial resolution taking time steps of an hour and more. Robert's model is based on two central ideas:

1) The grid points used in interpolating to the departure point of a particle are chosen so that they always surround the departure point; thus, when the winds are strong this set of grid points may be many grid intervals upstream from the arrival point of the particle (we use the term "multiply-upstream" to describe a scheme using interpolation points chosen in this way).

2) A semi-Lagrangian treatment of advection in the vorticity equation is combined with a semi-implicit treatment of the gravity wave terms in the divergence and continuity equations.

In the present paper, we clarify point 1) above by carrying out a stability analysis of some simple multiply-upstream semi-Lagrangian advective schemes, showing that they are unconditionally stable for a constant flow. We then extend the semi-Lagrangian approach to the horizontal advection terms of a multi-level, split explicit, primitive equation model and show that this allows a faster integration of the model than has previously been possible.

2. Stability analysis

We consider the advection equation

$$d\psi/dt = 0 \tag{1}$$

in one and two dimensions. Integrating over the tra-

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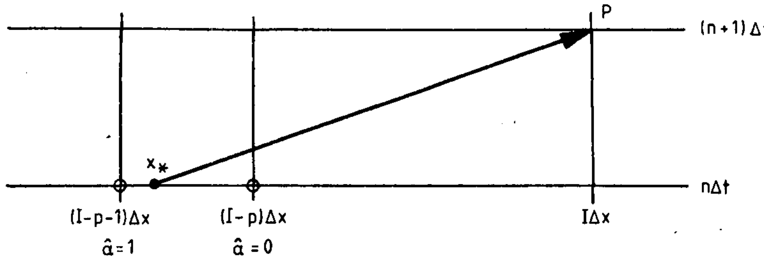

FIG. 1. The two interpolation points used in obtaining ψ_*^n by linear interpolation are circled.

jectory of a particle which arrives at a grid point P at time $(n + 1)\Delta t$ we have

$$\psi^{n+1}(P) = \psi_*^n, \tag{2}$$

where ψ_*^n is the value of ψ at the departure point of the particle at time $n\Delta t$. The value of ψ_*^n is obtained by polynomial interpolation from neighboring gridpoints, the stability and accuracy of the scheme depending on the type of interpolation used. We examine five cases of interpolation, all of which reproduce the exact solution at the interpolation gridpoints.

a. One-dimensional flow. Linear interpolation

In this case the situation is as shown in Fig. 1. A constant flow u (we assume $u > 0$ without loss of generality) advects a particle from its departure point x_* at time $n\Delta t$ to its arrival gridpoint P at time $(n + 1)\Delta t$. The grid interval within which the departure point lies is p grid intervals upstream from the arrival point ($p \geq 0$). We obtain ψ_*^n by linear interpolation from the surrounding gridpoints $(I - p)$ and $(I - p - 1)$. Hence (2) gives

$$\psi_I^{n+1} = (1 - \hat{\alpha})\psi_{I-p}^n + \hat{\alpha}\psi_{I-p-1}^n, \tag{3}$$

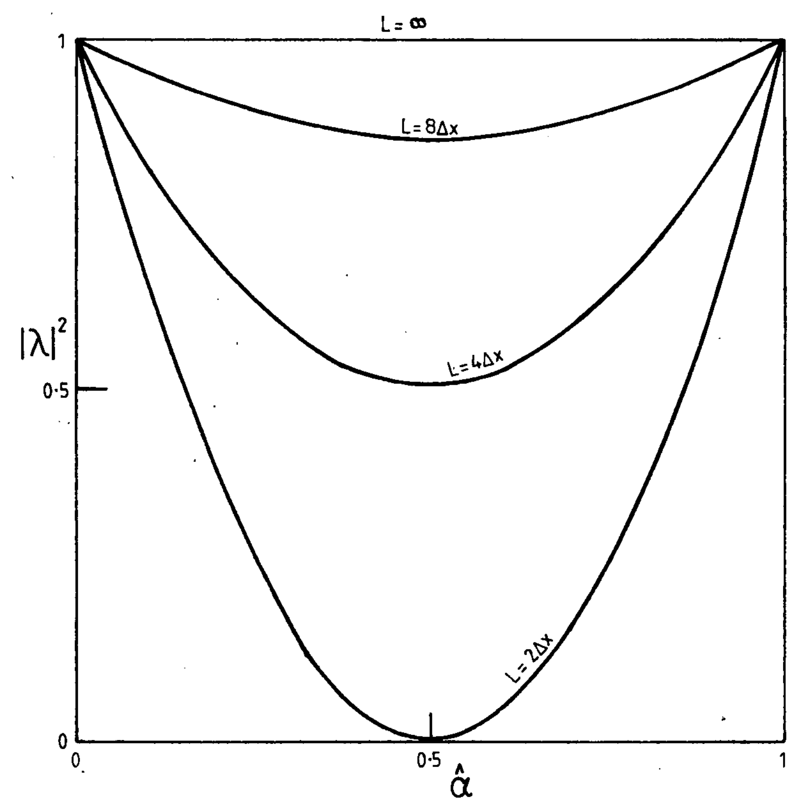

FIG. 2. Squared modulus of the amplification factor as a function of $\hat{\alpha}$ for various wavelengths (case of linear interpolation).

where

$$\hat{\alpha} = \alpha - p, \quad \alpha = u\Delta t/\Delta x.$$

Assuming a solution

$$\psi_I^n = \Psi^0 \lambda^n \exp(ikI\Delta x), \tag{4}$$

Eq. (3) gives

$$\lambda = \{1 - \hat{\alpha}[1 - \exp(-ik\Delta x)]\} \exp(-ipk\Delta x). \tag{5}$$

If we write

$$\lambda = |\lambda| \exp(-i\omega^* \Delta t),$$

we have

$$|\lambda|^2 = 1 - 2\hat{\alpha}(1 - \hat{\alpha})(1 - \cos k\Delta x), \tag{6}$$

$$\omega^* \Delta t = pk\Delta x + \tan^{-1}\left[\frac{\hat{\alpha} \sin k\Delta x}{1 - \hat{\alpha}(1 - \cos k\Delta x)}\right].$$

The scheme is stable ($|\lambda|^2 \leqslant 1$), provided

$$0 \leqslant \hat{\alpha} \leqslant 1, \tag{7}$$

i.e., provided the departure point lies within the interpolation interval $(I - p, I - p - 1)$. But we have chosen the interval so that this is the case; thus the scheme is unconditionally stable.

In Fig. 2, $|\lambda|^2$ is plotted as a function of $\hat{\alpha}$ for various wavelengths L. We see that heavy damping occurs for the shortest wavelengths (complete extinction of the shortest resolvable wave $L = 2\Delta x$ occurs for $\hat{\alpha} = 0.5$), but the damping gets less as the wavelength increases.

Since the analytical frequency ω is given by ku, the relative frequency R_1 ($=\omega^*/\omega$) is given by

$$R_1 = \frac{1}{(p + \hat{\alpha})k\Delta x} \times \left\{pk\Delta x + \tan^{-1}\left[\frac{\hat{\alpha} \sin k\Delta x}{1 - \hat{\alpha}(1 - \cos k\Delta x)}\right]\right\}.$$

In the long-wave limit ($k\Delta x \rightarrow 0$), we see that $R_1 \rightarrow 1$. Also, if the departure point coincides with a gridpoint ($\hat{\alpha} = 0$ or 1) we see that $R_1 = 1$. In Fig. 3(a), R_1 is plotted as a function of wavelength for $\hat{\alpha} = 0.25$. We see that $R_1 \rightarrow 1$ as p increases, i.e., the phase errors actually decrease as $u\Delta t/\Delta x$ exceeds unity. This result is, of course, peculiar to the case of a constant wind field, where the departure point can be precisely located using the wind at the arrival point.

b. One-dimensional flow: Quadratic interpolation

In this case the situation is as shown in Fig. 4. The gridpoint nearest the departure point x_* is chosen as the central point of the three interpolation points $(I - p + 1)$, $(I - p)$, $(I - p - 1)$. Thus x_* lies within a half grid interval from $(I - p)$. Using quadratic interpolation, (2) gives

$$\psi_I^{n+1} = 0.5\hat{\alpha}(1 + \hat{\alpha})\psi_{I-p-1}^n + (1 - \hat{\alpha})(1 + \hat{\alpha})\psi_{I-p}^n - 0.5\hat{\alpha}(1 - \hat{\alpha})\psi_{I-p+1}^n, \tag{8}$$

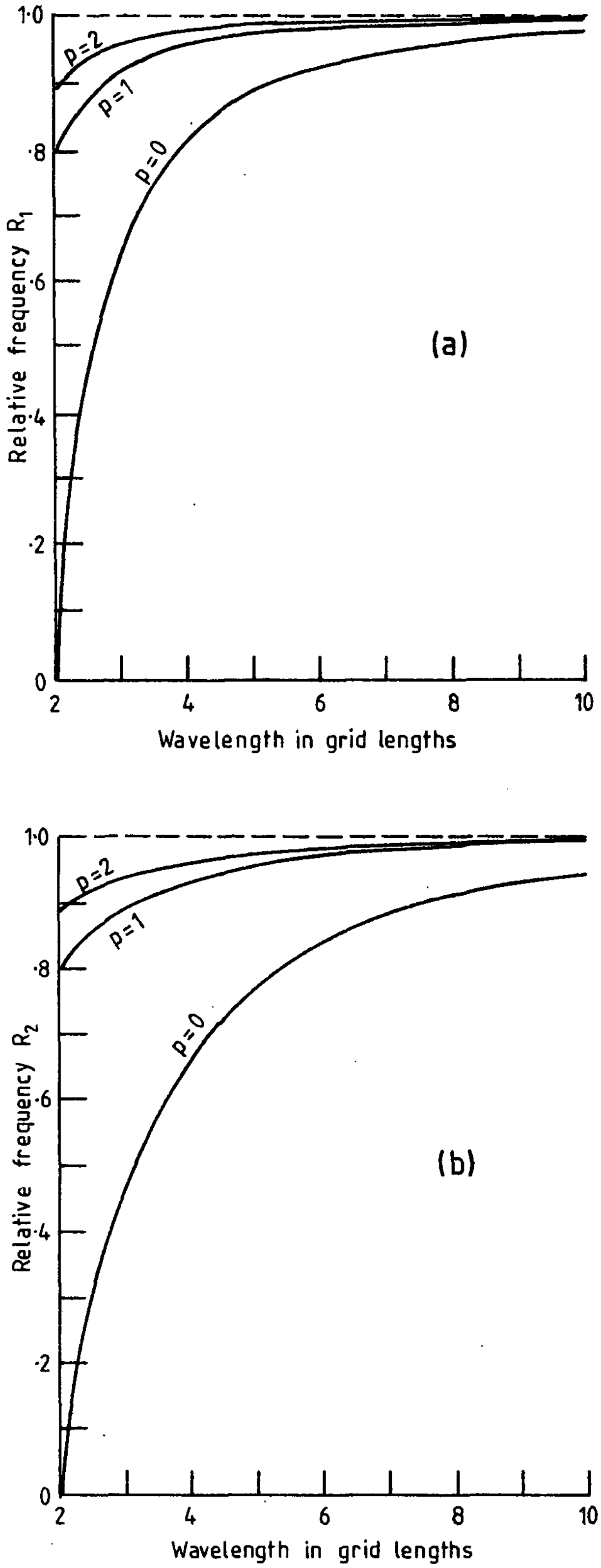

FIG. 3. Relative frequency as a function of wavelength in grid lengths for (a) the case of linear interpolation and (b) the case of quadratic interpolation, for various values of p with $\hat{\alpha} = 0.25$.

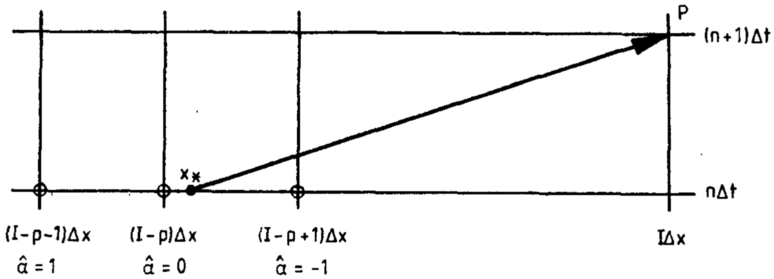

FIG. 4. The three interpolation points used in obtaining ψ_*^n by quadratic interpolation are circled.

and assuming a solution of the form (4) we have

$$\lambda = [1 - \hat{\alpha}^2(1 - \cos k\Delta x) - i\hat{\alpha}\sin k\Delta x]\exp(-ipk\Delta x). \quad (9)$$

Thus

$$\left.\begin{aligned} |\lambda|^2 &= 1 - \hat{\alpha}^2(1 - \hat{\alpha}^2)(1 - \cos k\Delta x)^2 \\ \omega^*\Delta t &= pk\Delta x + \tan^{-1}\left[\frac{\hat{\alpha}\sin k\Delta x}{1 - \hat{\alpha}^2(1 - \cos k\Delta x)}\right] \end{aligned}\right\}. \quad (10)$$

The scheme is stable provided

$$-1 \leqslant \hat{\alpha} \leqslant 1. \quad (11)$$

But our interpolation points are chosen such that

$$-0.5 \leqslant \hat{\alpha} \leqslant 0.5. \quad (12)$$

Thus, this scheme is again unconditionally stable.

In Fig. 5, $|\lambda|^2$ is plotted as a function of $\hat{\alpha}$ for various wavelengths L. Since we choose $\hat{\alpha}$ to lie in the interval defined by (12), we see that complete extinction never occurs and the damping is, in all cases, much less than that given by linear interpolation.

The relative frequency for the case of quadratic interpolation, denoted by R_2, is given by

$$R_2 = \frac{1}{(p + \hat{\alpha})k\Delta x} \times \left\{pk\Delta x + \tan^{-1}\left[\frac{\hat{\alpha}\sin k\Delta x}{1 - \hat{\alpha}^2(1 - \cos k\Delta x)}\right]\right\}.$$

Again, in the long-wave limit $R_2 \to 1$, and if the departure point coincides with a gridpoint $R_2 = 1$. In Fig. 3b, R_2 is plotted as a function of wavelength for $\hat{\alpha} = 0.25$. Again we see that $R_2 \to 1$ as p increases. Unlike the amplitude representation, we note that the phase representation is not improved by going to quadratic interpolation.

c. Two-dimensional flow: Bilinear interpolation

In this case the situation is as shown in Fig. 6. The wind components (u, v) are assumed constant and, without loss of generality, positive. Four interpolation points are chosen to surround the departure point (x_*, y_*). The interpolation box is (p, q) grid intervals upstream in the (x, y) directions from the arrival point $(p \geqslant 0, q \geqslant 0)$. Using bilinear interpolation (2) gives

$$\psi_{I,J}^{n+1} = (1 - \hat{\alpha})(1 - \hat{\beta})\psi_{I-p,J-q}^n + \hat{\alpha}(1 - \hat{\beta})\psi_{I-p-1,J-q}^n + \hat{\beta}(1 - \hat{\alpha})\psi_{I-p,J-q-1}^n + \hat{\alpha}\hat{\beta}\psi_{I-p-1,J-q-1}^n, \quad (13)$$

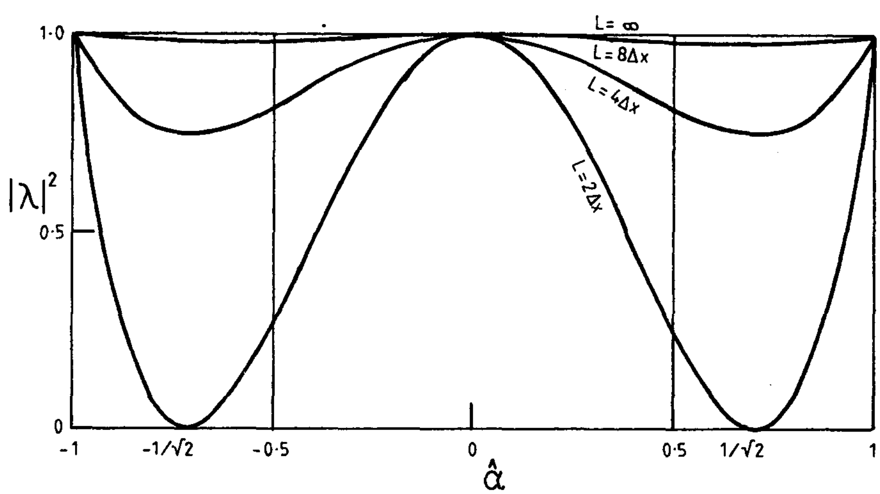

FIG. 5. As in Fig. 2, but for the case of quadratic interpolation.

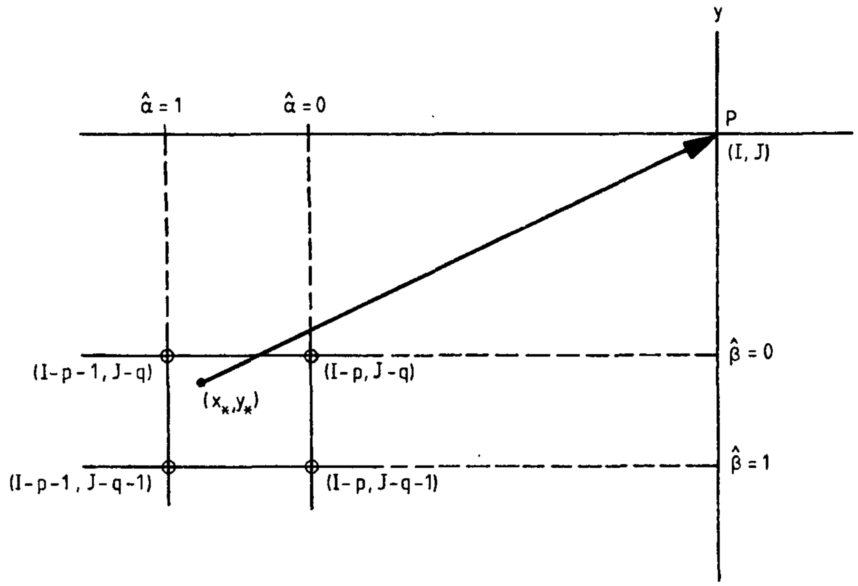

FIG. 6. The four interpolation points used obtaining ψ_*^n by bilinear interpolation are circled.

where $\hat{\alpha}$ is defined as before, and

$$\hat{\beta} = \beta - q, \quad \beta = v\Delta t/\Delta y.$$

Assuming a solution

$$\psi_{I,J}^n = \Psi^0\lambda^n \exp[i(kx + ly)], \tag{14}$$

Eq. (13) gives

$$\lambda = \lambda_1\lambda_2,$$

where

$$\lambda_1 = \{1 - \hat{\alpha}[1 - \exp(-ik\Delta x)]\} \exp(-ipk\Delta x), \tag{15}$$

$$\lambda_2 = \{1 - \hat{\beta}[1 - \exp(-il\Delta y)]\} \exp(-iql\Delta y), \tag{16}$$

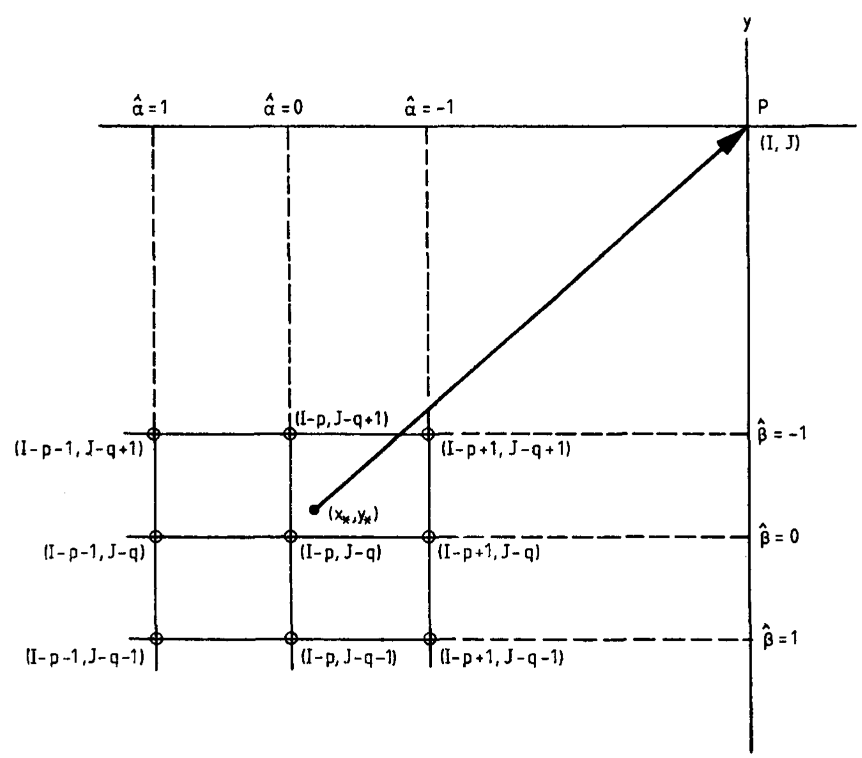

FIG. 7. The nine interpolation points used in obtaining ψ_*^n
by biquadratic interpolation are circled.

i.e., the amplification factor is the product of two factors which are of the same form as in the one-dimensional case of Section 2a. Thus we have the theoretical simplicity of a splitting method, even though no splitting has been used.

A sufficient condition for stability is

$$(0 \le \hat{\alpha} \le 1) \quad \text{and} \quad (0 \le \hat{\beta} \le 1), \tag{17}$$

which holds if (x_*, y_*) lies within the interpolation box. But this is guaranteed by our initial choice of interpolation points. Thus, the scheme is unconditionally stable.

As in the case of Section 2a, heavy damping occurs for the shortest resolvable waves.

d. Two-dimensional flow: Biquadratic interpolation

In this case the situation is as shown in Fig. 7. The grid-point nearest the departure point is chosen as the center of the nine points used in the interpolation. The interpolation formula used is the Lagrangian formula (e.g. Carnahan *et al.*, 1969, p. 65).

$$\psi^n(x, y) = \sum_{\substack{i=-1\\j=-1}}^{\substack{j=1\\i=1}} W_{i,j}\psi^n_{i,j}, \tag{18}$$

where the (i, i) range over the nine interpolation points, and

$$W_{i,j} = \prod_{\substack{k=-1\\k\neq i}}^{k=1} \frac{(x - x_k)}{(x_i - x_k)} \prod_{\substack{l=-1\\l\neq j}}^{l=1} \frac{(y - y_l)}{(y_j - y_l)}.$$

This formula reproduces the exact solution at the gridpoints, and when applied to (2) gives

$$\begin{aligned}\psi^{n+1}_{I,J} = {} & 0.25\hat{\alpha}(1 + \hat{\alpha})\hat{\beta}(1 + \hat{\beta})\psi^n_{I-p-1,J-q-1} + 0.5(1 - \hat{\alpha}^2)\hat{\beta}(1 + \hat{\beta})\psi^n_{I-p,J-q-1} - 0.25\hat{\alpha}(1 - \hat{\alpha})\hat{\beta}(1 + \hat{\beta})\psi^n_{I-p+1,J-q-1} \\ & + 0.5\hat{\alpha}(1 + \hat{\alpha})(1 - \hat{\beta}^2)\psi^n_{I-p-1,J-q} + (1 - \hat{\alpha}^2)(1 - \hat{\beta}^2)\psi^n_{I-p,J-q} - 0.5\hat{\alpha}(1 - \hat{\alpha})(1 - \hat{\beta}^2)\psi^n_{I-p+1,J-q} - 0.25\hat{\alpha} \\ & \times (1 + \hat{\alpha})\hat{\beta}(1 - \hat{\beta})\psi^n_{I-p-1,J-q+1} - 0.5(1 - \hat{\alpha}^2)\hat{\beta}(1 - \hat{\beta})\psi^n_{I-p,J-q+1} + 0.25\hat{\alpha}(1 - \hat{\alpha})\hat{\beta}(1 - \hat{\beta})\psi^n_{I-p+1,J-q+1}.\end{aligned} \tag{19}$$

Assuming a solution of the form (14), we then find

$$\lambda = \lambda_1\lambda_2,$$

where

$$\lambda_1 = [1 - \hat{\alpha}^2(1 - \cos k\Delta x) - i\hat{\alpha}\sin k\Delta x] \times \exp(-ipk\Delta x)$$

$$\lambda_2 = [1 - \hat{\beta}^2(1 - \cos l\Delta y) - i\hat{\beta}\sin l\Delta y]\exp(-iql\Delta y),$$

i.e., the amplification factor is the product of two factors which are of the same form as in the one-dimensional case (b), so that again we have the theoretical simplicity of a splitting method, without doing any splitting.

A sufficient condition for stability is

$$(-1 \le \hat{\alpha} \le 1) \quad \text{and} \quad (-1 \le \hat{\beta} \le 1). \tag{20}$$

But our initial choice of interpolation points is such that

$$(-0.5 \le \hat{\alpha} \le 0.5) \quad \text{and} \quad (-0.5 \le \hat{\beta} \le 0.5). \tag{21}$$

Thus the scheme is unconditionally stable.

As in case (b), complete extinction never occurs and the damping is in all cases much less than that given by bilinear interpolation.

e. Two-dimensional flow in an E-grid

The *E*-grid (see Mesinger and Arakawa, 1976) is a semi-staggered grid in which vector and scalar quantities are expressed at alternate gridpoints. We consider the semi-Lagrangian approach in such a grid for a constant wind field. Bilinear and biquadratic interpolation are most easily defined and analysed if we adopt the axes (X, Y), in general oblique, which run through grid points of a given type (see Fig. 8). Let the wind components in the oblique axes be (u_E, v_E). Then defining

$$\hat{\alpha} = \alpha - p, \quad \alpha = u_E\Delta t/\Delta X,$$

$$\hat{\beta} = \beta - q, \quad \beta = v_E\Delta t/\Delta Y,$$

we use bilinear interpolation in the (X, Y) coordinates to obtain from (2) an expression of exactly the same form as (13). Assuming a solution

$$\psi^n_{I,J} = \Psi^0\lambda^n \exp[i(k_E X + l_E Y)],$$

we find

$$\lambda = \lambda_1\lambda_2,$$

where

$$\lambda_1 = \{1 - \hat{\alpha}[1 - \exp(-ik_E\Delta X)]\}\exp(-ipk_E\Delta X),$$

$$\lambda_2 = \{1 - \hat{\beta}[1 - \exp(-il_E\Delta Y)]\}\exp(-iql_E\Delta Y).$$

Thus again a sufficient condition for stability is

$$(0 \le \hat{\alpha} \le 1) \quad \text{and} \quad (0 \le \hat{\beta} \le 1), \tag{22}$$

which is guaranteed by our choice of interpolation box.

Similarly, the analysis for the case of biquadratic interpolation in the (X, Y) coordinates proceeds as in (d), giving formally identical results.

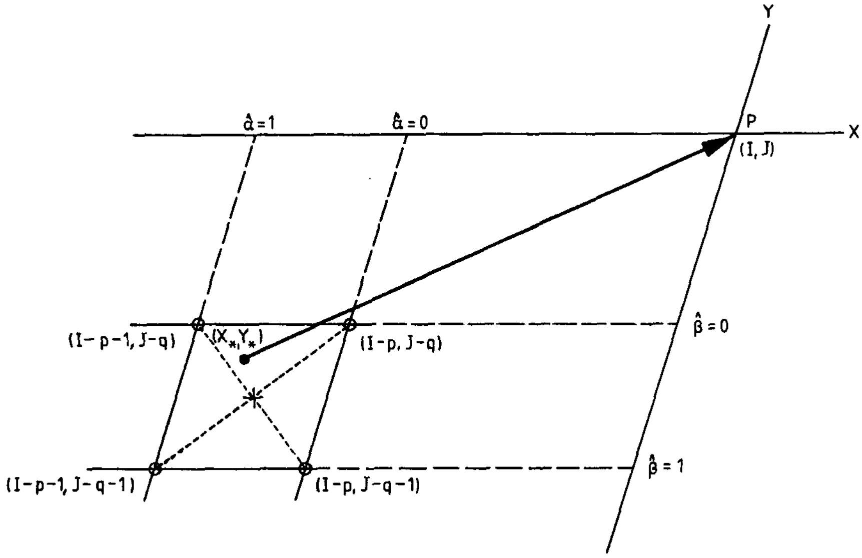

FIG. 8. The four interpolation points used in obtaining ψ_*^n by bilinear interpolation in the oblique axes (X, Y) are circled. The direction of the original orthogonal axes are shown by the dotted lines.

3. Application to a multi-level model

In this section, we describe the results of computations in which the semi-Lagrangian approach has been applied to the horizontal advection terms of the NWP model used operationally in the Irish Meteorological Service (IMS). Operational forecasts made with the model in its Eulerian form serve as our reference with which to compare the results of the semi-Lagrangian integrations.

a. The Eulerian model

The model is a modified version of the HIBU model, developed at the Federal Hydrometeorological Institute and Belgrade University, Yugoslavia (Janjić, 1977, 1979; Mesinger, 1977, 1981). It is a limited area, split explicit, primitive equation model on an *E*-grid (Arakawa notation), using latitude–longitude coordinates and the sigma vertical coordinate. The horizontal advection terms are expressed in an energy and enstrophy conserving form, various options being provided for the time-integration of these terms. A forward–backward time scheme is used to integrate the gravity wave terms, while the trapezoidal implicit scheme, with the time step, is used to integrate the Coriolis terms. The model includes a treatment of the divergence term in the continuity equation which prevents the accumulation of two-grid-interval noise. Only very simple physics are employed, moisture and radiation being omitted. There is a simple treatment of surface friction and vertical eddy momentum transport. Dry convective adjustment is applied once every eight time-steps.

The principal modification made to the HIBU model in introducing it into operational use in the Irish Meteorological Service was to rotate the latitude–longitude coordinate system so that the pole was displaced into the Pacific (it now lies at 150°E, 30°N). This gives a much more uniform grid in our area of interest and saves considerably on computer time, since polar filtering is no longer necessary. (The stratagem of rotating a model's spherical coordinate system had previously been used at the Swedish Meteorological and Hydrological Institute; Undén, 1980).

Our integration area, shown in Figs. 9–13, is spanned by 81×26 gridpoints carrying the same variable, located at alternate intersections of a $1° \times 1°$ horizontal mesh. Thus the grid distance, defined as the shortest distance between similar points, is 157 km at the model equator. Five levels are used in the vertical, the three upper layers having $\Delta\sigma = 0.25$ and the two lower layers having $\Delta\sigma = 0.125$. The model top is at 200 mb. The time step Δt is 7.5 min, this being imposed by the stability criterion for gravity-inertia waves.

We update the variables on the two outermost boundary lines of the model every time step, using values derived from previous predictions of the European Centre for Medium-Range Weather Forecasts. On the next three lines in from the boundary the integrations are performed using a simple upwind difference scheme for horizontal advection, which gives high damping. In addition, light second-degree diffusive damping is applied on the five outermost boundary lines. Divergence damping (Sadourny, 1975), formulated so as to have no effect on the rotational part of the flow, is applied over the whole area for the full period of the integration, with a coefficient of 1.3×10^6 m^2 s^{-1}.

The model as described above was used operation-

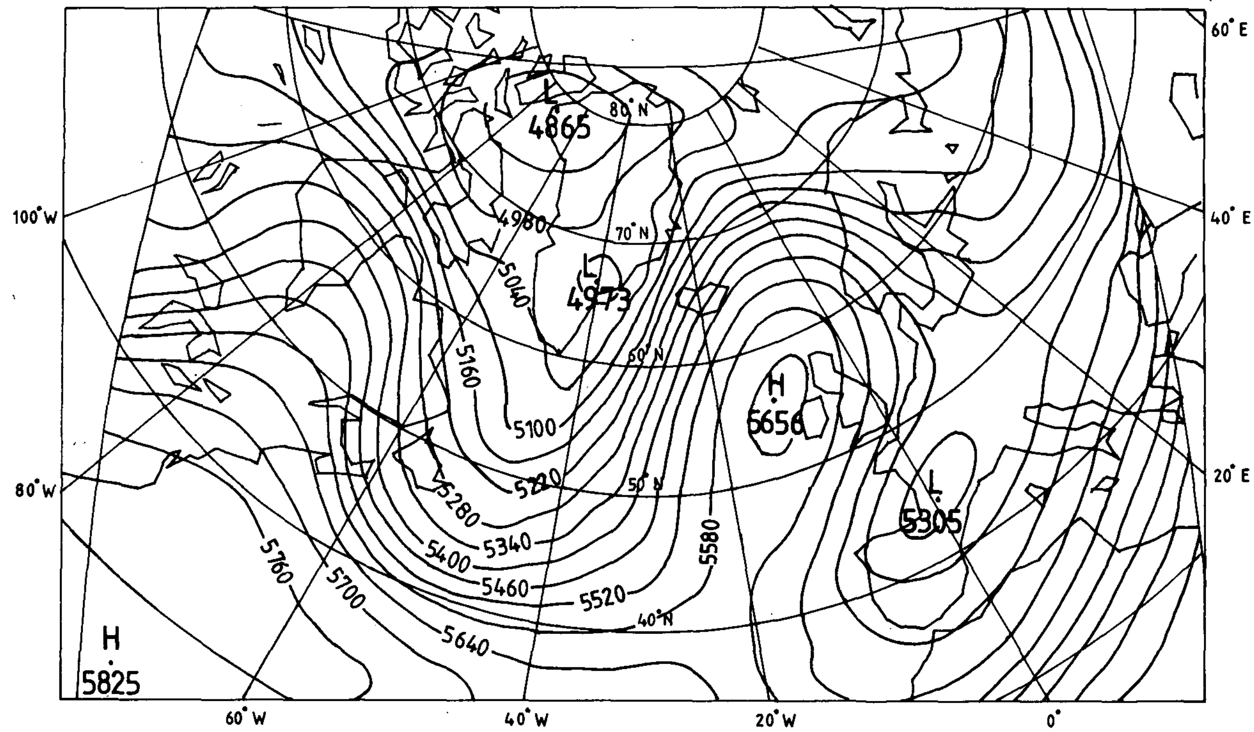

FIG. 9. 24 h 500 mb forecast for 31 March 1982 at 0000 GMT, performed with the Eulerian model. Time step 7.5 min for all terms. Contours in gpm.

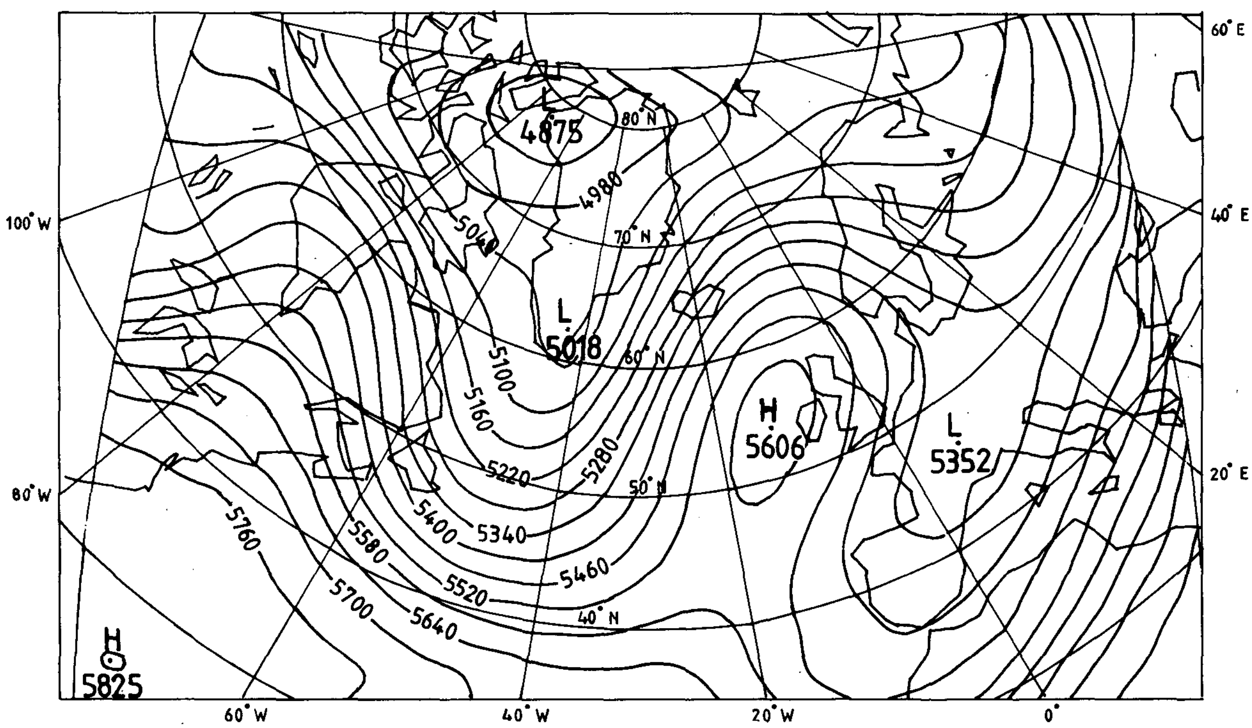

FIG. 10. As in Fig. 9, but performed with the model having semi-Lagrangian advection. Case of bilinear interpolation. Adjustment time step 7.5 min, advection time step 30 min.

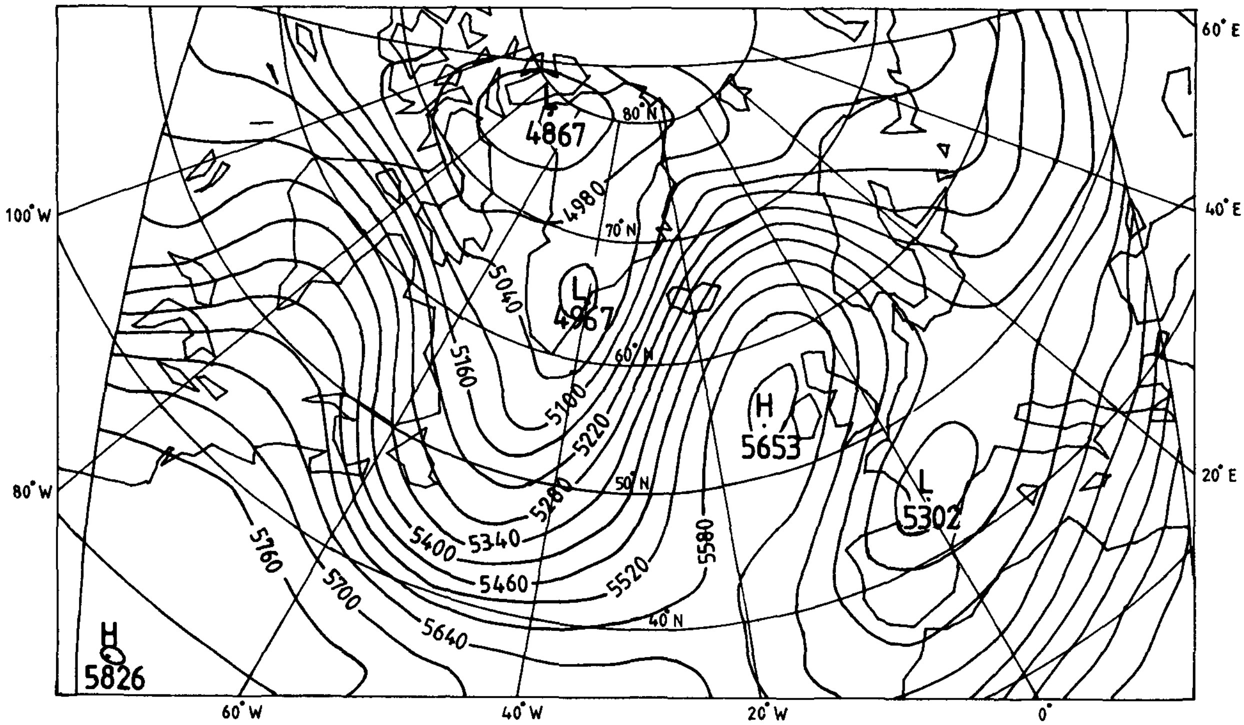

FIG. 11. As in Fig. 10, but for case of bi-quadratic interpolation.

ally between July 1980 and May 1982 to produce 36-h forecasts twice a day on a DEC 20/50 computer having a speed of approximately 1 MIPS and a core storage of 256 K words. The model resides in core. Using an Adams-Bashforth time scheme for the horizontal advection terms, which was found most sat-

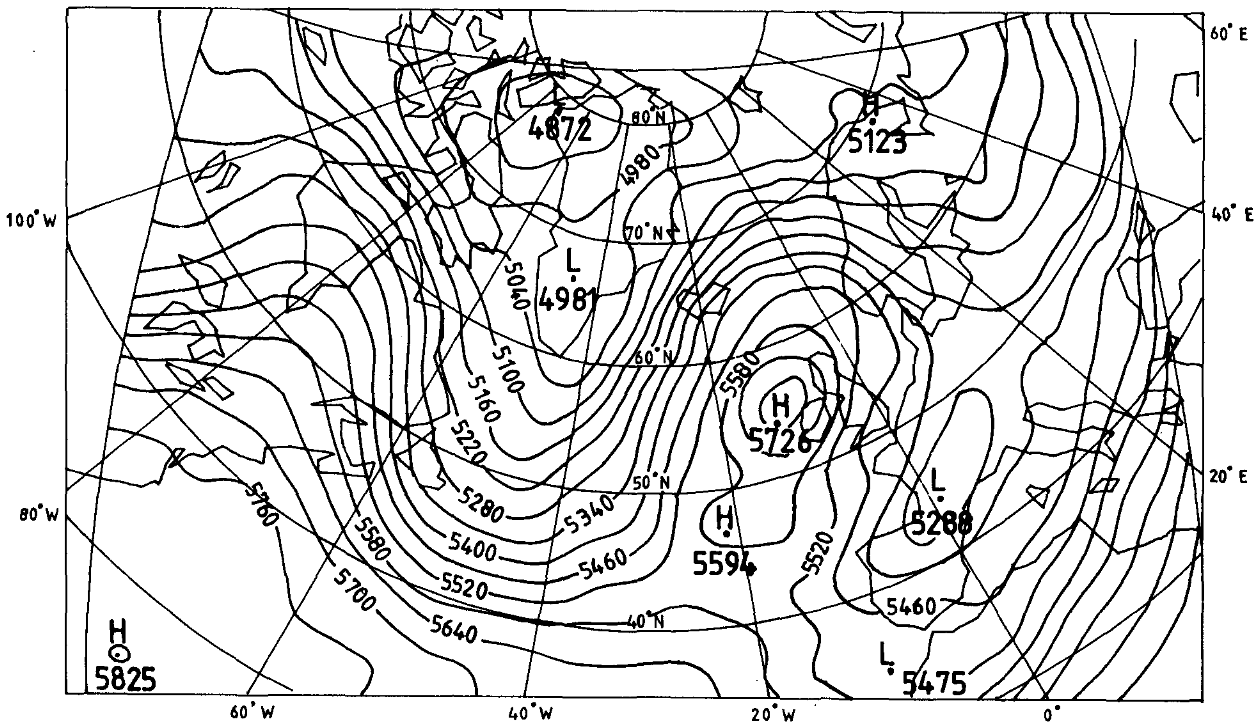

FIG. 12. As in Fig. 11, but with an advection time step of 1 h.

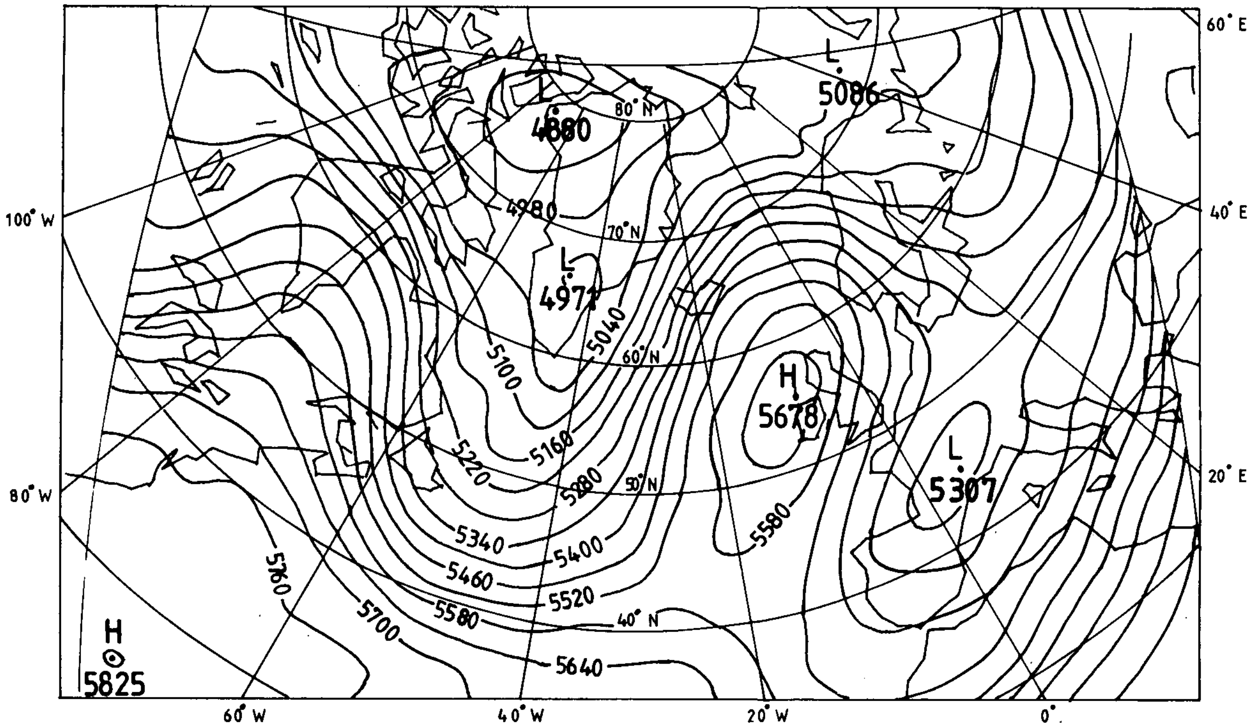

FIG. 13. As in Fig. 12, but with position of departure points estimated using an iterative procedure.

isfactory in operational practice, the CPU time required for a 24 h integration was 37 min, of which 17 min was spent on the horizontal advection.

b. Attempts to use multiple time steps for advection in the Eulerian model

In a model based on the splitting method, the separate phases of the integration have their own stability criteria. In principle, therefore, it is possible to economize on computer time by using a multiple time step for the advective terms, since these have a more lenient stability criterion than do the adjustment terms. We have made numerous attempts to use this theoretical advantage of the splitting method in the framework of our Eulerian model, but without success.

With the single step Adams-Bashforth or Euler-forward time schemes for horizontal advection, we have found it possible to ensure stability only by using the same time step Δt for advection as is used for adjustment. With the two step Heun scheme, we have been able to achieve a time step of $2\Delta t$ for advection. However, there is no net saving of computer time in this case, since the additional intermediate calculations required by the Heun scheme cancel the advantage of using a double time step.

As far as we are aware, no other users of split explicit Eulerian models have been able to use multiple time steps for advection with a single step time scheme. Marchuk (1974), in his textbook devoted mainly to the splitting method, does not refer to any such application. Janic (personal communication, 1980) and others (Gadd, 1978; Duffy, 1981) have succeeded in using $3\Delta t$ for advection, but in all cases in combination with a two-step time scheme requiring intermediate calculations.

c. The model with semi-Lagrangian advection

We now modify the Eulerian model by taking the horizontal advection terms in the momentum and thermodynamic equations and expressing them in the form $(du/dt)_H = 0$, $(dv/dt)_H = 0$, $(dT/dt)_H = 0$. We integrate these terms in the semi-Lagrangian manner, chosing a multiple time step $N\Delta t$. The remaining terms are kept in their original Eulerian form and integrated as before with the single time step Δt.

On arriving at a time $n\Delta t$ in the cycle of integration when a horizontal advection step is to be performed, the departure points of particles which will arrive at the gridpoints (I, J) at time $(n + N)\Delta t$ are first estimated. This is done by taking the available wind components $(u, v)^n_{I,J}$, regarding them as valid at $(n + N/2)\Delta t$, and using them to go back from the gridpoints over the interval $N\Delta t$ to find the departure points at time $n\Delta t$. (A time truncation error enters here, of course; it can be reduced by adopting an iterative procedure—see below.) Gridpoints surrounding the departure points are chosen as interpolation points, in the manner described in Section 2. The advective integration is then performed according to the formula

$$\psi_{I,J}^{n+N} = \psi_*^n, \tag{23}$$

where ψ_*^n is evaluated by bilinear or biquadratic interpolation in the (X, Y) coordinates of the E-grid, as described in Subsection 2*e*.

The above description refers to the integration of the momentum equations. In the case of the thermodynamic equation, where wind components are not available at the relevant gridpoints, the trajectories are estimated using winds obtained by four-point spatial averaging.

The advective step is followed by N applications of the adjustment step, acting in the first instance on the values $\psi_{I,J}^{n+N}$ obtained from (23).

Numerical experiments have been performed with the model thus modified, for the cases of both bilinear and biquadratic interpolation. All non-advective parameters have been kept at the same values as in the Eulerian model. With the semi-Lagrangian approach we have found it possible, for the first time, to use multiple time steps for advection without having to employ an intermediate time stepping procedure within this phase.

The integrations for the bilinear case, though very efficient and completely stable, have been found to give undesirably smooth forecast charts; this is in accord with the analysis of Section 2. We show in Fig. 9 a 24 h 500 mb forecast performed with our Eulerian model. The corresponding bilinear semi-Lagrangian forecast with an advective time step of $4\Delta t$ is shown in Fig. 10. Though the patterns are similar, it can be seen that considerable smoothing has taken place. The CPU time spent on the horizontal advection phase, however, has been reduced from 17 to 3 min for the 24 h integration, representing an overall saving of 38% (the time taken from the adjustment phase remains unchanged).

The results for the case of biquadratic interpolation are much more satisfactory. We show in Fig. 11 the corresponding forecast for this case, again using $4\Delta t$ for advection. Comparing this with the Eulerian forecast, we see that the results are very close. The time now taken for the advective phase is 4 min for the 24 h integration, representing an overall saving in CPU time of 35% compared to the Eulerian run.

In this biquadratic semi-Lagrangian run, the advective time step was large enough so that 201 vector points and 188 scalar points (mostly at the top level) had p or q greater than zero at the beginning of the integration. After 24 h, 157 vector points and 145 scalar points still had p or q greater than zero. Thus, the multiply-upstream interpolation was being fully used.

Sixty 36 h forecasts were run in parallel over the period 6 April–8 May 1982 to compare the biquadratic semi-Lagrangian model as described above with the Eulerian model. In all cases, the integrations remained stable, and subjective comparison showed no significant differences between the two sets of forecasts. Some objective verification statistics derived from these runs are shown in Table 1. No loss of accuracy is indicated on going over to the semi-Lagrangian model, despite its longer advective time step.

TABLE 1. Verification statistics for the Eulerian model, and the biquadratic semi-Lagrangian model with a time step of $4\Delta t$ for advection, for sixty 24 h (36 h) forecasts. The statistics were evaluated on a central verification area.

	Eulerian model	Biquadratic semi-Lagrangian model
rms error (m) in 500 mb geopotential	34.37 (48.90)	32.78 (46.90)
Standard error in the mean (m) in 500 mb geopotential	1.29 (1.94)	1.18 (1.82)
rms error (mb) in surface pressure	3.72 (5.12)	3.70 (5.15)
Standard error in the mean (mb) in surface pressure	0.15 (0.20)	0.15 (0.19)

As a result of these tests, the biquadratic semi-Lagrangian model with the time step of $4\Delta t$ for advection has been adopted for operational use in the IMS, beginning on 10 May 1982. In addition to economy in computer time, the semi-Lagrangian model has the advantage of requiring less computer storage than the Eulerian model. This is due to the fact that the Eulerian model with the Adams-Bashforth time scheme for advection required storage of the three-dimensional arrays relating to the advection of (u, v, T) at the previous as well as the current time level.

To further test the properties of the biquadratic semi-Lagrangian model, we have performed a number of 36 h integrations using an advective time step of $8\Delta t$. The integrations have all remained stable, though localized noise is evident on some of the forecast charts. As an example, the 24 h forecast corresponding to Fig. 9 is shown in Fig. 12. In addition to the appearance of noise, there has been a noticeable loss of accuracy compared to the Eulerian forecast. In the present case, 1949 vector points and 1954 scalar points had p or q greater than zero at the beginning of the integration, while the corresponding figures after 24 h are 1749 and 1725, respectively. Extensive use was therefore being made of the multiply-upstream interpolation. Our subjective impression is that the noise arises in regions where significant changes in wind speed or direction occur.

In an effort to reduce the noise and increase the accuracy of the integration having an $8\Delta t$ advective time step, we have performed some experiments in which the departure points are estimated using an iterative procedure. As a first step in the iteration, we proceed as before, using $(u, v)_{I,J}^n$ to estimate the departure points of particles which arrive at the gridpoints (I, J) at time $(n + 8)\Delta t$. A first biquadratic semi-Lagrangian update is performed and the values $(u, v)_{I,J}^{n+8}$ obtained are regarded as provisional. An average of the velocity components $(u, v)_{I,J}^{n}$ and $(u, v)_{I,J}^{n+8}$ is then taken, and the departure points are reestimated using these average values. This is followed by a second biquadratic semi-Lagrangian update, again acting on the time level n quantities, giving the new values $(u, v, T)_{I,J}^{n+8}$. The results of a 24 h integration using this iterative procedure, valid for the same time as Fig. 9, are shown in Fig. 13. The forecast is indeed more accurate (relative to the Eulerian forecast) and less noisy than Fig. 12, but not as accurate or as noisefree as Fig. 11, where a $4\Delta t$ advective time step was used without iteration. The CPU time required by the iterative run was almost the same as that required for the forecast of Fig. 11. For a given expenditure of computer time, therefore, it appears preferable to use shorter time steps without iteration, rather than longer time steps with iteration.

Finally we note that Robert (1981) has used bicubic interpolation in his integrations, which he finds to be expensive. Our results would seem to indicate that the biquadratic is as high an order of interpolation as is required.

4. Conclusions

An analysis of the stability properties of some simple semi-Lagrangian advective schemes has shown that it is possible to circumvent the usual Courant-Friedrichs-Lewy condition for stability by adopting a multiply-upstream approach. This consists of choosing gridpoints surrounding the departure points of the particles at the beginning of a time step for interpolating the values of advected variables. This procedure is in accord with the general principle governing stability, which requires that the domain of dependence of the numerical solution at a gridpoint should surround the physical characteristic passing through that point.

The semi-Lagrangian approach has been applied to a split explicit primitive equation model with simple physics, allowing the advective terms to be integrated with a multiple time step. Taking a half-hour time step for advection, the new approach saves more than one-third of the computer time required to integrate the model in Eulerian form, while causing no perceptible loss of accuracy.

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